



Safety Risk Assessment of Existing Sluices in the Tarim River Basin Based on Analytic Hierarchy Process



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Abstract: Strengthening or rebuilding dangerous sluices to restore their functions is an urgent issue calling for attention in order to guarantee the safety of people's lives and properties. Based on analytic hierarchy process (AHP), this study analyzed in detail the influences of flood control capacity, seepage, structural and seismic safety, metal structure, electromechanical equipment, and engineering quality on the comprehensive evaluation of sluice gates. It also established a safety evaluation index system for the gates in service in the Tarim River Basin, and applied it to the safety evaluation of 25 sluice gates. The degree of importance of each sluice was quantified by the index of sluice building level and the design of water diversion flow; the calculation method of sluice risk index was established by combining the importance and the safety indices of sluices. The study demonstrated that the safety ranking of 25 sluice gates could corroborate with the safety appraisal results. The ranking of the urgency of derisking were more reasonable and in line with actual situations. This proposed method is simple, practical, and operable to scientifically evaluate the safety and risks of existing sluices, hence exhibiting considerable engineering value for the consolidation and sequencing of dangerous sluices pending for reinforcement.

Keywords: Sluice gates; Safety evaluation; Analytic hierarchy process; Risk index; Tarim River Basin

1 Introduction

In the 20th century, a number of domestic sluices were built with concrete and metal structures approaching or even exceeding their service lives. However, at the time of construction, design and construction technologies were rudimentary, standards were low, and maintenance was insufficient. Though in operation, several sluices have developed serious defects, creating significant safety hazards [1]. According to the existing data, in China there are 1,782 large and medium-sized sluices, including 260 large and 1,522 medium-sized sluices. Dangerous and large-sized sluices account for 53.5% of the total number of large sluices; dangerous and medium-sized sluices account for 46.4% of the total number of medium sluices [2]. In 2009, there were 291 dangerous sluices, including 27 large and 264 medium-sized sluices, in Xinjiang as listed in the National Overall Plan for Reinforcement of Large and Medium-sized Dangerous Sluices. Currently, it is common for large and medium-sized sluices in Xinjiang to operate even if they face problems. As of 2021, 28 dangerous sluices in the plan were reinforced. The remaining 253 sluices (including 69 dangerous sluices) in the plan have not yet been reinforced, but are still operating despite their problems [3, 4]. This considerably affects the operation safety of the sluice projects.

One of the critical points of domestic water conservancy projects is the reinforcement or rebuilding of the existing dangerous sluices. However, the vital premise of reinforcing or rebuilding sluices aims at the formulation of a correct and comprehensive scientific safety evaluation plan. According to the requirements laid down in Measures for Safety Appraisal of Sluices, safety appraisal of domestic sluices should be conducted regularly at 5 to 10-year intervals [5]. At present, the safety appraisal of sluices is based on the standard in Guide for Safety Evaluation of Sluices. The process includes the collection of the sluice project data, completion of the safety appraisal or evaluation, and classification of the qualitative safety evaluation of sluices. However, this method could neither quantitatively compare the safety of sluices in the same safety category, nor compare the risks of sluices of different engineering grades. It could not effectively support the sorting of sluice reinforcement. In the research on sluices, Zhang and Tang [6] studied the theoretical assessment method of structural reliability of the aging states of sluice gates. Ma et al. [7] assumed that the deformation of various parts of sluice includes both overall and individual

effects. By applying the maximum entropy principle to the probability distribution function of individual effect extremes, it effectively identified the overall deformation trend of sluice and the degree of deviation of measured values from the overall deformation, thereby completing the evaluation and analysis of non-uniform deformation states. Xue et al. [8] employed neural network technology to conduct dimensionality reduction research on the safety evaluation index system of complex hydraulic gates. Based on the monitoring data of the middle orifice gate of Shaping Second-cascade Hydropower Station, a comparative verification was carried out on the results before and after dimensionality reduction of the evaluation index system. Although this method demonstrated good performance in deformation prediction and anomaly identification, the internal mechanism of the model was difficult to interpret intuitively, and the connection between the output results and the physical process was not sufficiently clear. Liu et al. [9] examined the aging assessment index, assessment index system, and assessment methods of sluice gates with district irrigation sluice gates as the research object. Sun [10] proposed the application of a comprehensive evaluation method for the safety evaluation of sluice gates. Zheng et al. [11] used the grey relational analysis method to evaluate sluices based on the specific conditions and management principles of dangerous sluices. Targeting to address the existing problems found in sluice safety evaluation, Li [12] adopted a new way of fault mode analysis. Chang and He [13] specified various evaluation indices that affected the reliability of sluice system based on the reliability evaluation model of the sluice system in the lower reaches of the Yellow River, and thoroughly considered them from three aspects: Safety, applicability, and durability. Accordingly, he specified the scoring criteria and values of relevant indexes. Zhang et al. [14] and Zhang and Yan [15] applied the improved analytic hierarchy process (AHP), time-varying reliability theory, and risk rate thresholds based on the analysis of the structural characteristics and operation modes of hydropower projects; they then established a project operation safety risk rate assessment model. Using the improved Diamond Search (D-S) algorithm and Back Propagation Neural Network (BPNN), a multisource data fusion method was proposed as the method was proven to effectively evaluate the safety status of buildings through multilevel data fusion. Sanaras et al. [16], Hammerling et al. [17, 18] and Zamarr et al. [19] used AHP to evaluate the technical status and operational risks of small hydraulic structures, such as sluices, dams, and others.

Several approaches have been applied to sluice safety evaluation, including AHP, grey relational analysis, time-varying reliability theory, and D-S evidence fusion. However, each has notable limitations when used for reinforcement prioritization. AHP and the maximum entropy principle are systematic but rely on subjective expert judgments; grey relational analysis works with small samples but is sensitive to data quality and lacks stability in ranking; time-varying reliability and D-S fusion offer high accuracy but require extensive monitoring data and complex computations, making them less practical for large-scale application. Moreover, none of these methods quantitatively incorporate the engineering importance of sluices (e.g., design of flow and building level) into risk ranking, nor can they differentiate urgency within the same safety class. This study proposed a risk index that combined safety scores (derived from AHP-based multi-criteria evaluation) with sluice importance quantified by building level and design of diversion flow. The framework adhered to Chinese safety appraisal specifications while being simple, data-efficient, and suitable for prioritization. This system was based on AHP, which not only adhered to the specifications, but also combined well with the sluice safety appraisal data. It possessed engineering universality, and could objectively and completely evaluate the safety of sluices. Based on safety evaluations, the degree of importance of each sluice was represented quantitatively by two sluice building level indices and the design of diversion flow. The risk evaluation index system of sluice in the Tarim River Basin was constructed by combining the safety and importance of the sluice. The model was applied to the risk evaluation in this study area, and the practicability and effectiveness of the model were then tested. The evaluation results provided a reference for the decision making of sluice project reinforcement and sequencing.

2 Methods

2.1 Construction of Sluice Safety Evaluation Index System

Owing to the steep slope and rapid flow, short flow, and high-sediment content of the river tributaries in the Tarim River Basin in Xinjiang, the following problems are common in sluice gates:

1. Wear and tear, erosion, difficulty starting gates, blockage, etc.
2. Sluice gates are susceptible to damage caused by ice loads and freezing in some areas due to severe weather conditions and ice damage.
3. Because most sluice foundations are composed of liquefiable soils, they are prone to strength loss and deformation during earthquakes, which can cause severe structural damage.

Besides the geographically specific problems mentioned above, sluices in the Tarim River Basin face basin-wide issues, including the following: confusing flood control standards, inadequate flood discharge capacity, structural seismic capacity incompliant with the specification requirements, serious erosion damage of energy dissipation facilities, aging mechanical and electrical equipment, lack of safety monitoring facilities, and unstable river potential.

As shown in Figure 1, this study proposed a thorough safety evaluation index system for sluices by referring to

the Guide for Safety Evaluation of Sluices (SL 214 2015) [20] and the collected sluice safety appraisal data, as well as considering the characteristics of sluices in the Tarim River Basin. The system included three different levels, namely, target, criterion, and index levels. Represented by A, B, and C, respectively, the system adhered closely to the specifications, such as the guideline layer B and refined indicator layer C, but also integrated well with the collection of sluice gate information. Based on AHP, the safety of sluice could be evaluated objectively and comprehensively.

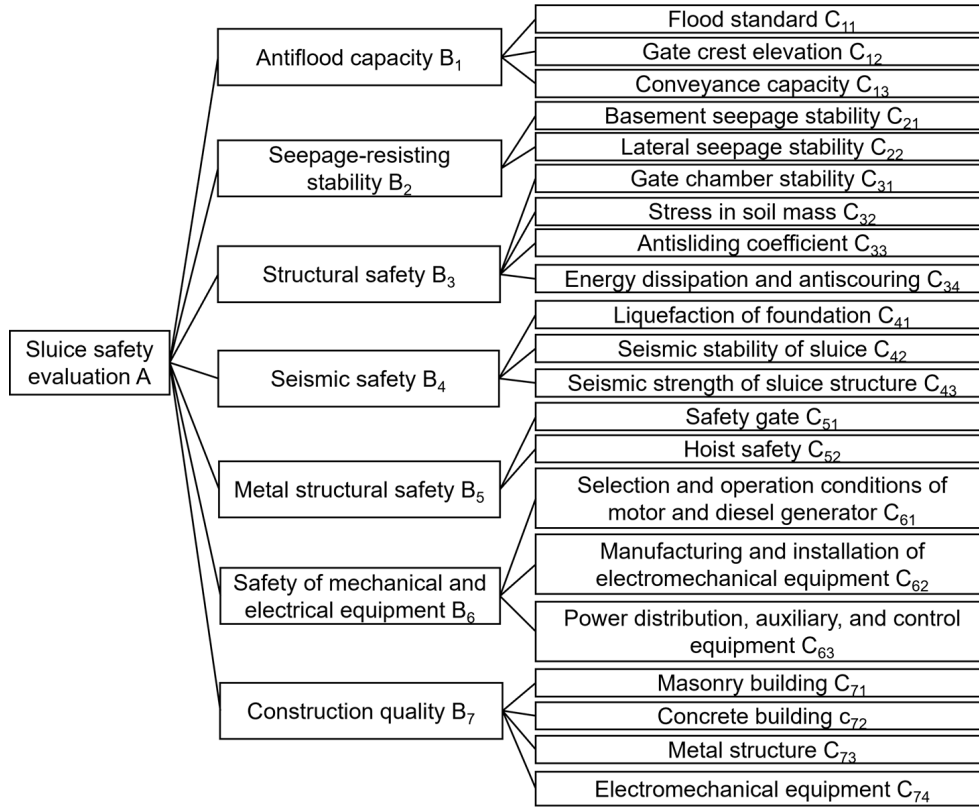


Figure 1. Evaluation index system

2.2 Weight Calculation Based on Analytic Hierarchy Process

AHP is an evaluation method that studies decision-making problems at different levels, combines quantitative and qualitative research, and constructs a judgment matrix to obtain the weight [3]. The steps of AHP analysis are as follows:

2.2.1 Establishment of hierarchical structure

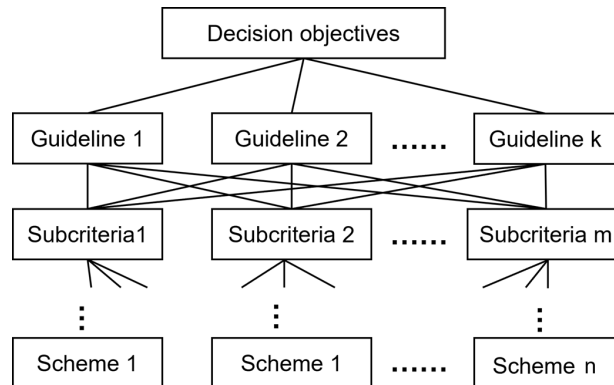


Figure 2. Hierarchical structure diagram

The goal of the decision-making problem was decomposed into several elements according to relevant attributes,

and these elements could be further decomposed into different levels. These levels could be divided into three categories, as below.

A. Highest level: Known as the target level. It has only one element, which is the goal to be achieved by the decision-making problem or the final result.

B. Middle layer: Known as the criterion layer. This layer is the intermediate link, which is dominated by the upper layer. It can also be decomposed into several layers which dominate the elements of the next layer.

C. Lowest level: Referred to as the measure or the scheme level. This level is the smallest element of the hierarchical structure and cannot be decomposed. The complexity of the decision-making problem determines the number of layers of hierarchy; as the complexity of the problem increases, the number of constructed layers increases. There is no limit on the number of layers of hierarchy. When there are too many criteria, sub-criteria layers should be decomposed further. The hierarchical structure diagram in Figure 2 clearly expresses the relationship among the levels.

2.2.2 Construction of judgment matrix

Quantifying specific gravity of each element is a critical problem associated with the construction of the judgment matrix when comparing the features in a standard layer. In addition, there may be many factors that influence a factor, and consideration of the extent to which this factor affects an aspect may be interfered with by a variety of other factors, poor judgment, incorrect data, and ultimately inconsistent results [3]. Compare n indicators, $A_1, A_2, \dots, A_n$, to determine the importance of criterion Z , so that the method of discriminating matrix could be used to evaluate the importance of n indicators. For Guideline Z , indicator A_i and the importance of indicator A_j indicated that n compared indicators constituting two comparative discriminating matrices, as shown in Table 1. The A_{ij} values were determined by numerical scaling, and were described in the most commonly used scale of 1–9. Table 2 lists the meanings of scales 1 to 9.

Table 1. Comparison discriminant matrix

Z	A_1	A_2	$\dots$	A_n
A_1	A_{11}	A_{12}	$\dots$	A_{1n}
A_2	A_{21}	A_{22}	$\dots$	A_{2n}
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
A_n	A_{n1}	A_{n2}	$\dots$	A_{nn}

Table 2. Scale meanings

Scale	Meaning
1	The two factors are equally important
3	Compared with the two factors, the former is slightly more important than the latter
5	Compared with the two factors, the former is obviously more important than the latter
7	Compared with the two factors, the former is more important than the latter
9	Compared with the two factors, the former is more important than the latter
2, 4, 6, 8	Indicates the intermediate value of the adjacent comparisons listed above

2.2.3 Hierarchical sorting

Hierarchical sorting involved the calculation of the eigenvalues of the judgment matrix first, and the determination of the eigenvector corresponding to the largest eigenvalue.

2.2.4 Consistency inspection

The judgment matrix constructed by the above method could effectively reduce the influences of other factors and reflect the importance of the two factors. Owing to the effect of human subjective factors, the solution of the judgment matrix scales would be inconsistent. Meanwhile, the following formula was adopted to test the consistency of the judgment matrix,

$$CR = \frac{CI}{RI} \quad (1)$$

$$CI = \frac{\lambda_{\max} - m}{m - 1} \quad (2)$$

where, CR is the random consistency ratio of the judgment matrix, CI is the general consistency index of the judgment matrix, and RI is the average random consistency index of the judgment matrix. The values of RI are listed in Table 3.

When $CR < 0.1$, the judgment matrix is close to the consistency, and the weight coefficients can be allocated reasonably. When this requirement cannot be met, it is necessary to return to the judgment matrix, and perform accurate correction until the consistency requirement can be met.

Table 3. Average random consistency index (RI) values

m	3	4	5	6	7	8	9
RI	0.58	0.90	1.12	1.24	1.32	1.41	1.45

Table 4. Index score value basis

Index	Normal [8,10]	Available lesions [5,7]	Invalidation [0,4]
Flood standard C_{11}	Engineering classification, building level, and flood standard all meet the requirements of current codes	Meet the requirements of current specifications, but the requirements of recent planning are not met	Engineering classification, building level and flood standard do not meet the requirements of current codes
Gate crest elevation C_{12}	Meet the current specification requirements	Does not meet the requirements of current codes, but can be solved by engineering measures	Does not meet the requirements of the current specification, and needs to be scrapped and rebuilt
Conveyance capacity C_{13}	Meet the design requirements	Meet the design requirements, but the requirements of recent planning are not met, or there is siltation	Does not meet the design requirements
Basement seepage stability C_{21}			
Lateral seepage stability C_{22}			
Gate chamber stability C_{31}			
Stress in soil mass C_{32}			
Antisliding coefficient C_{33}			
Energy dissipation and anticouring C_{34}	Meet the requirements of current specifications and operate normally	Meet the requirements of the current specification, and the quality defects still do not affect the overall safety	Does not meet the requirements of the current specification, or cannot operate normally
Liquefaction of foundation C_{41}			
Seismic stability of sluice C_{42}			
Seismic strength of sluice structure C_{43}			
Safety gate C_{51}			
Hoist safety C_{52}			
Selection and operation conditions of motor and diesel generator C_{61}			
Manufacturing and installation of electromechanical equipment C_{62}			

Index	Normal [8,10]	Available lesions [5,7]	Invalidation [0,4]
Power distribution equipment, auxiliary equipment and control equipment C ₆₃			
Masonry building C ₇₁	Meet the current specification requirements; no obvious erosion, wear, freeze–thaw, and other defects are found in the operation, and the current situation meets the operation requirements	This basically meets the requirements of current codes, and obvious erosion, abrasion, freeze–thaw and other defects found in operation still do not affect the safety of the project	Most of them do not meet the requirements of current codes, or the aforementioned serious quality problems are encountered in the operation of the project, which affects the safety of the project
Concrete building C ₇₂			
Metal structure C ₇₃	Meet the current specification requirements, no quality defects are found in operation, and the current situation meets the operation requirements	Meet the basic requirements of the current specification, and the quality defects found in the operation will not affect the engineering safety	Most of them do not meet the requirements of the current specifications, or quality problems have been found in the operation of the project, which affects the safety of the project
Electromechanical equipment C ₇₄			

2.3 Safety Index Score Criteria

In general terms, both quantitative and qualitative indicators exist in the safety evaluation index system. There are three ways to obtain quantitative index values: a) by consulting relevant data; b) based on formula calculations; and c) based on actual measurements. Data cannot be used directly to calculate the qualitative index value, but the evaluation value can only be obtained by empirical processing and quantification based on the complete understanding of the evaluation object, mainly achieved by expert consultation. As this evaluation was based on the data of 25 sluice gates in Xinjiang, which were mostly safety appraisal data, most indicators could not be used in quantitative form. Therefore, a qualitative method was used and the dimensionless approach was adopted to deal with the indicators. Through the scoring criteria and expert consultation, the indicators were scored in the same interval and the values reflected corresponding grades. In this study, the break of [0, 10] was adopted and the three districts were divided into three corresponding grading levels. Each index was scored according to the following standards, and the values of index scoring standards are listed in Table 4.

2.4 Calculation Method of Risk Indicators

In practice, it is often impossible to complete strengthening work simultaneously because many sluices in a specific area need strengthening. In particular, the risk rate of sluice gates in Xinjiang is high, and many sluice gates associated with problems are still operating. Strengthening works are long-term jobs to be conducted in batches lasting for many years. Establishing risk assessment index ranking calculation method of sluice gates could distinguish priorities, prioritize the risky diseased reservoirs for strengthening and derisking, arrange the derisking and strengthening plan scientifically and reasonably, and provide decision makers with a scientific and logical basis for derisking and strengthening implementation [21]. Therefore, it is necessary to sort out the urgency of derisking, and to strengthen sluices in conjunction with the actual project.

Based on safety evaluations, this study quantified the importance of each sluice with the index derived from the design of diversion flow. It tried to combine the safety and importance of sluices to obtain the respective risk indexes, which could be used as the ranking basis of the urgency of sluice reinforcement, that is, as the risk of sluice increased, the urgency of sluice reinforcement increased.

Firstly, the value based on the design of diversion flow was converted into the importance index value according to the following formula [22]:

$$a_i = b + 0.8 \frac{D_i}{D_{\max}} \quad (3)$$

where, i = the engineering scale grade, a_i = the value of importance index, and b = the constant related to i , if $i = 4$, $b = 1$; if $i = 3$, $b = 2$, and if $i = 2$, $b = 3$. In addition, D_i = design of diversion flow value for sluice, and $D_{\max}$ = maximum design of diversion flow value for sluice at the corresponding scale level.

The importance index value of each sluice can be obtained by Eq. (3), and the importance score can then be obtained according to the following equation [23]:

$$E = (10 - C)a_i \quad (4)$$

where, E = risk score value, and C = score for each sluice safety evaluation.

In summary, the risk evaluation index system of sluice in the Tarim River Basin was established in this study, and its structural framework is shown in Figure 3.

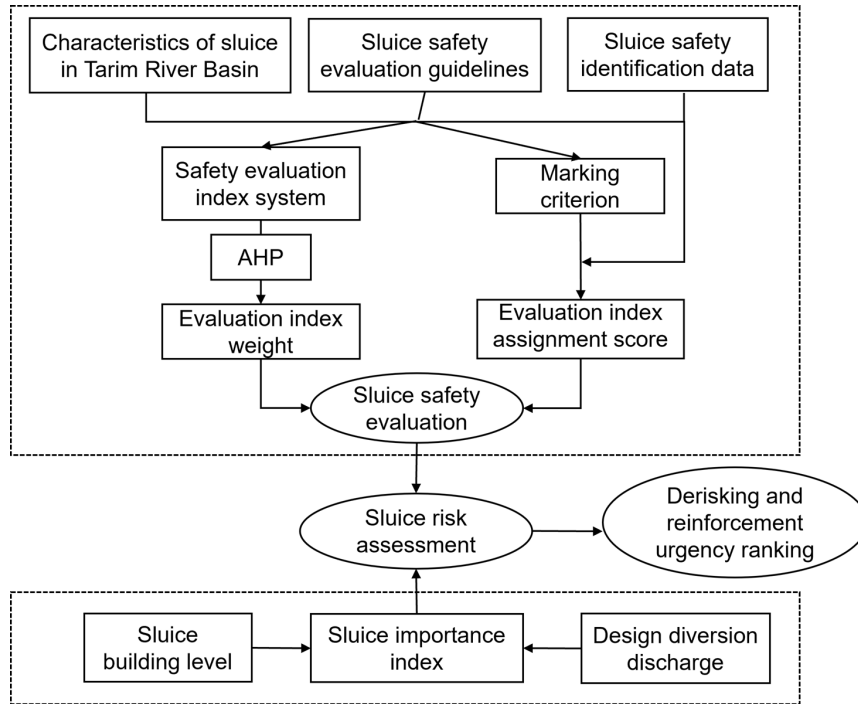


Figure 3. Index system of sluice risk assessment in the Tarim River Basin

3 Applied Research

3.1 Project Overview

There were 25 sluices in the Tarim River Basin which could be used for safety evaluation. Among these, thirteen were located in the Hotan area, nine in the Bazhou area, and three in the Aksu area. Table 5 lists the details of sluices including their names, specific locations, and project scales. The distribution of each sluice is shown in Figure 4.

Table 5. Sluice projects in Tarim River Basin

Numbering	Name of Sluice	Location	Scale
1	Urastai River Xiebinebhu Canal Intake Sluice	Bazhou Prefecture and Hejing County	III
2	Laolongkou Water Diversion Hub of Qiang River	Xinlongkou, Keping County, Aksu Prefecture	III
3	Kuchman Canal Head, Yutian County	Yutian County, Hotan	IV
4	Moyu County Flood Diversion Gate	Mo Yuxian in Hotan area	II
5	Ayitak Canal Head Project in Minfeng County	Yeyike Township, Minfeng County, Hotan	III
6	Moyu County River Gate	Mo Yuxian in Hotan area	II
7	Yangxia River Water Control Project	Bazhou Luntai County	III
8	First Gate of Awati County, Laoda River	Ayikule town, Aksu city	III
9	Moyu County Qushou Hydropower Station Gate	Mo Yuxian in Hotan area	II
10	Moyu County Gate	Mo Yuxian in Hotan area	II
11	Aqia Canal Head	Keping County, Aksu Prefecture	III

Numbering	Name of Sluice	Location	Scale
12	Kunur Water Diversion Project	Bazhou Luntai County	III
13	Dina River Diversion Hub	Bazhou Luntai County	III
14	Tariq River Diversion Hub	Luntai County, Bazhou.	III
15	Cedaya River Diversion Hub	Cedaya town, Luntai County, Bazhou	III
16	Aqimak Sluice, Moyu County	Mo Yuxian in Hotan area	II
17	Dinahe Old Water Diversion Sand Sluice Diversion Hub	Bazhou Luntai County	III
18	Yutian County Liberation Canal Head	Yutian County, Hotan Prefecture	III
19	Yutian County Unity Channel Head	Yingbage Township, Yutian County, Hotan	III
20	Yeyungou Water Diversion Project	Bazhou luntai County	III
21	Dina River Old Flood Drainage Gate Diversion Hub	Bazhou luntai County	III
22	Head of the Sayivak River, Cele County	Sayiwake Village, Qile County, Hotan	IV
23	Qiaha River Head of Shengli Reservoir in Cele County	Qiaha Township, Qia County, Hotan	III
24	Kumutuge Gate	Hotan city	III
25	Diversion gate of Miligawati Hydropower Station	Hotan city	III

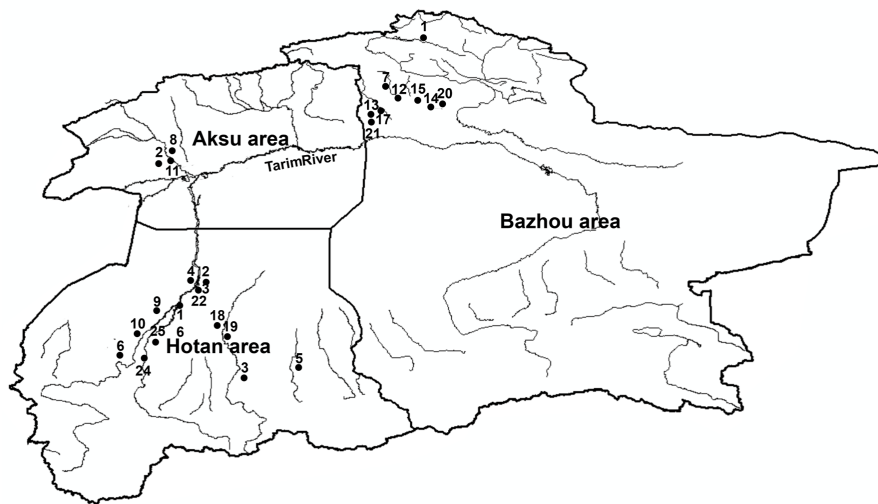


Figure 4. Sluice distribution map

Table 6. Percentage of 25 sluices with various problems

Problem Type	Proportion
Inadequate flood control capacity	44.80%
Insufficient antipermeability stability	31.05%
Insufficient structural safety	44.80%
Insufficient seismic capacity	34.50%
Insufficient safety of metal structure	58.60%
Insufficient safety of mechanical and electrical equipment	48.30%
Poor engineering quality	75.90%

According to the statistics of the collected data on sluice gates, the proportions of the 25 sluice gates with various types of problems are listed in Table 6. As indicated, multiple types of problems are prevalent in these sluice gates in the Tarim River Basin. Up to 75.9% are associated with poor engineering quality evaluations, and they do not

meet the specification standards. At least 30% of the sluice gates have other problems. In this regard, there is an urgent need for safety evaluation and risk assessment.

3.2 Results of Safety Evaluation

According to the results of AHP calculations, the weights of each index are listed in Table 7.

Table 7. Calculation results of weight of each index

Target Layer	Criteria Layer	Weight	Index Layer	Weight
A	B ₁	0.3965	C ₁₁	0.2511
			C ₁₂	0.0421
			C ₁₃	0.1033
	B ₂	0.2014	C ₂₁	0.1510
			C ₂₂	0.05034
			C ₃₁	0.02634
	B ₃	0.1488	C ₃₂	0.02634
			C ₃₃	0.06043
			C ₃₄	0.03564
	B ₄	0.06335	C ₄₁	0.02112
			C ₄₂	0.02112
			C ₄₃	0.02112
	B ₅	0.06335	C ₅₁	0.01584
			C ₅₂	0.04751
	B ₆	0.06335	C ₆₁	0.02112
			C ₆₂	0.02112
			C ₆₃	0.02112
	B ₇	0.06335	C ₇₁	0.01927
			C ₇₂	0.02455
			C ₇₃	0.01135
			C ₇₄	0.00818

The results of the matrix consistency test are listed in Table 8. The listings in the Table indicated that the matrix was close to consistency, and the weight coefficient distribution was reasonable.

Table 8. Matrix consistency test results

Matrix	A	B ₁	B ₂	B ₃	B ₄	B ₅	B ₆	B ₇
Consistency Ratio	0.0309	0.0334	-	0.0569	0	-	0	0.0575

The specific scores yielded by each evaluation index were weighted, and the final evaluation result was obtained. First, the overall safety evaluation of these 25 sluices was sorted, as shown in Table 9.

The listings in Table 9 show that the overall safety evaluation ranking of 25 sluices could be confirmed by the safety appraisal results. The top six sluices were all Class II (In the Safety Evaluation Guidelines for Sluices, Class I sluice is the best safety identification whereas Class IV sluice is the lowest), while the bottom sluices were Class IV. However, some were abrupt, such as Cedaya River Diversion Hub and Aqimake Sluices, which ranked 15th and 16th, respectively. Their safety appraisal results were Class III, but their rankings were all among Class IV sluices. According to the safety appraisal, Cedaya River Diversion Hub was a Class III sluice; the Pier's concrete of the flood discharge and sand washing sluice in this project was defective, cracked, and aged. The bottom plate of sluice was seriously worn, the bracket was cracked and exposed, and the structure had quality defects. Although the flood-discharge sluice, diversion sluice, and overflow weir were anti-sliding and stable at various working conditions, the foundation stress response met the allowable requirements of the code. The measures of energy dissipation and scour prevention did not meet the criteria (the flood-discharge sluice had no stilling basin in its original design). It was doubtful whether its structural safety was rated as Grade B. In addition, the seismic safety, metal structural safety, mechanical and safety of electrical equipment, and other aspects of the project did not meet the code's requirements, all of which were classified as Grade C. Even though the sluice was rated as the third-class sluice according to the code, it was the lowest standard of the third-class sluice. In this study, AHP, a scoring method that thoroughly considered different levels and weights, was used to evaluate the safety of the project so the total safety score of the project was low. Similar to Cedaya River Diversion Hub, the Aqimak Sluice in Moyu County was of Grade B in

terms of flood control standard, structural and seepage safety. Its engineering quality, anti-seismic, metal structure, and mechanical and electrical equipment were all ranked Grade C, which was the lowest standard for Grade III gates. Moreover, the sluice's New Yuejin Canal Control Gate and Old Yuejin Canal Branch Gate could not meet the use requirements. Therefore, its safety should be rated as Grade C according to the standard flood control standard, and should belong to Grade IV gates. The lowest evaluation was the head of Kuchman Canal Head. According to the safety appraisal data of this sluice, the seven safety indices were all Grade C and the score was thus the lowest.

Table 9. Safety evaluation sluice results

Ranking	Name of Sluice	Safety Grade	Appraisal Result of Sluice
1	Dina River Old Flood Drainage Gate Diversion Hub	8.22	Class II sluice
2	Qiaha River Head of Shengli Reservoir in Cele County	8.17	Class II sluice
3	Water Diversion Gate of Mirigawati Hydropower Station in Hotan City	8.13	Class II sluice
4	Head of the Sayivak River	8.12	Class II sluice
5	Yeyungou Water Diversion Project	8.08	Class II sluice
6	Kumutuge Gate, Hotan City	7.85	Class II sluice
7	Yutian County Liberation Canal Head	7.65	Class III sluice
8	Yutian County Unity Channel Head	7.53	Class III sluice
9	Tariq River Diversion Hub	7.21	Class IV sluice
10	Dinahe Old Water Diversion Sand Sluice Diversion Hub	7.14	Class IV sluice
11	First Gate of Awati County, Laoda River	6.67	Class IV sluice
12	Dina River Diversion Hub	6.67	Class IV sluice
13	Moyu County Gate	6.62	Class IV sluice
14	Aqia Canal Head	6.57	Class IV sluice
15	Cedaya River Diversion Hub	6.41	Class III sluice
16	Aqimak Sluice	6.24	Class III sluice
17	Moyu County Qushou Hydropower Station Gate	6.09	Class IV sluice
18	Kunur Water Diversion Project	5.76	Class IV sluice
19	Urastai River Xiebinebhu Canal Intake Sluice	5.01	Class IV sluice
20	Yangxia River Water Control Project	5.00	Class IV sluice
21	Moyu County River Gate	4.66	Class IV sluice
22	Moyu County Flood Diversion Gate	4.13	Class IV sluice
23	Laolongkou Water Diversion Hub of Qiang River	4.03	Class IV sluice
24	Ayitak Canal Head Project in Minfeng County	4.01	Class IV sluice
25	Kuchman Canal Head	3.97	Class IV sluice

3.3 Sorting Results of Danger Elimination

Following calculations, the orders of criticality of danger removal and reinforcement of the 25 studied sluices are listed in Table 10.

The listings in Table 10 show that the top four sluices in danger removal and reinforcement are of type II, which was attributed to their high importance and low safety score. Compared with the 6th and 7th sluices, the scale of Moyu County Gate was type II, which was more critical than the Laolongkou Water Diversion Hub of Qiang River. Still, its safety grade was much higher than the latter, while its risk was lower than the latter. Although Kuchman Canal Head, ranked 8th, was a IV project with minor importance, its safety score ranked second from the last; accordingly, the comprehensive risk was high. The urgency of danger removal and reinforcement was also high. The head of the Sayivak River ranked lowest in the urgency of danger removal and reinforcement. This was because this project scale was IV, with low importance and a high-safety score. Thus, the risk was the lowest, and the urgency of danger removal and reinforcement ranked last.

Table 10. Order of criticality of danger removal and reinforcement

Ranking	Name of Sluice	Safety Grade	Safety Appraisal	Significance	Risk	Scale
1	Moyu County Flood Diversion Gate	4.13	Class IV sluice	3.80	22.29	II
2	Moyu County River Gate	4.66	Class IV sluice	3.50	18.70	II

Ranking	Name of Sluice	Safety Grade	Safety Appraisal	Significance	Risk	Scale
3	Moyu County Qushou Hydropower Station Gate	6.09	Class IV sluice	3.71	14.49	II
4	Aqimak Sluice	6.24	Class III sluice	3.60	13.52	II
5	Ayitak Canal Head Project in Minfeng County	4.01	Class IV sluice	2.12	12.70	III
6	Laolongkou Water Diversion Hub of Qiang River	4.03	Class IV sluice	2.10	12.53	III
7	Moyu County Gate	6.62	Class IV sluice	3.60	12.17	II
8	Kuchman Canal Head, Yutian County	3.97	Class IV sluice	1.80	10.85	IV
9	Yangxia River Water Control Project	5.00	Class IV sluice	2.12	10.59	III
10	Urastai River Xiebinebhu Canal Intake Sluice	5.01	Class IV sluice	2.12	10.58	III
11	Dina River Diversion Hub	6.67	Class IV sluice	2.80	9.32	III
12	Kunur Water Diversion Project	5.76	Class IV sluice	2.04	8.65	III
13	First Gate of Awati County, Laoda River	6.68	Class IV sluice	2.60	8.64	III
14	Cedaya River Diversion Hub	6.41	Class III sluice	2.04	7.32	III
15	Aqia Canal Head	6.57	Class IV sluice	2.08	7.13	III
16	Dinahe old water diversion sand sluice diversion hub	7.14	Class IV sluice	2.16	6.17	III
17	Tariq River Diversion Hub	7.21	Class IV sluice	2.08	5.80	III
18	Yutian County Unity Channel Head	7.53	Class III sluice	2.15	5.31	III
19	Yutian County Liberation Canal Head	7.65	Class III sluice	2.20	5.17	III
20	Kumutuge Gate	7.85	Class II sluice	2.22	4.77	III
21	Water Diversion Gate of Mirigawati Hydropower Station	8.13	Class II sluice	2.54	4.76	III
22	Qiaha River Head of Shengli Reservoir in Cele County	8.17	Class II sluice	2.31	4.22	III
23	Yeyungou Water Diversion Project	8.08	Class II sluice	2.08	3.99	III
24	Dina River Old Flood Drainage Gate Diversion Hub	8.22	Class II sluice	2.16	3.84	III
25	Head of the Sayivak River	8.13	Class II sluice	1.53	2.88	IV

4 Results

Constructing a sluice evaluation index system is relatively trivial but complicated. Based on AHP, reference to the specifications and data, in conjunction with the characteristics of the sluice in Tarim River Basin, the analyses of problems related to sluices were done in accordance with the principles of constructing the evaluation index system. This study summarized the comprehensive evaluation index system of sluice safety and provided the specific index scoring standard. Finally, the safety evaluation and sequencing of the 25 sluices in the Tarim River Basin were conducted. The analysis showed that the overall safety ranking of the 25 sluices studied here could confirm the safety appraisal results, and findings were in line with actual situations. This showed that the safety evaluation results of 25 sluices obtained by AHP were reasonable and accurate.

Based on the safety evaluation, the importance of each sluice was quantified by the index of sluice building level and design of diversion flow. By mathematical means, the risk index of the sluice was obtained (which served as the basis for the urgency ranking of sluice reinforcement), and the emergence urgency ranking of 25 sluices was achieved. The results of the urgency ranking of sluice reinforcement are in line with reality, which proves that the risk calculation method proposed in this study is reasonable. This method can evaluate scientifically the safety of existing sluices and has significant engineering application value for the consolidation and sequencing of dangerous sluices.

5 Discussion

Compared with other studies of the same kind, the novelty of this paper lies in establishing an alternative system for the safety evaluation and risk assessment of sluice gates. This system is close to the specification and can be closely related to the domestic safety appraisal work, so it assists in the tedious and daunting task of data collection. The superiority lies in the simplicity of the method without losing the science and the practical application of engineering. However, since the research object in this paper was the Tarim River Basin sluice gates, the evaluation indexes and criteria took into account the characteristics of the Tarim River Basin sluice gates. For other research objects, appropriate adjustments need to be made based on this paper, having considered the characteristics of regional sluices rather than being applied directly.

6 Conclusions

This study developed a safety and risk assessment framework for existing sluices in the Tarim River Basin based on the AHP. The main findings and contributions are summarized as follows:

1. The proposed index system based on the AHP encompassed seven criteria, including flood control, leakage, structural safety, seismic resistance, metal structure, electromechanical equipment, and engineering quality, along with 22 indicators. It successfully ranked 25 sluices. The ranking results align with the official findings from safety assessment, thus affirming the effectiveness of this method.

2. By combining safety scores with the importance of gates, we derived a risk index. The resulting urgency ranking prioritized gates with lower safety scores, while those with higher safety scores were ranked in descending order. This result aligns with engineering intuition and provides a more effective identification tool than relying solely on safety classification level. The proposed framework was simple, data-parsimonious, and directly usable by water conservancy agencies for scheduling sluice reinforcement. It could reduce the arbitrariness in deciding which sluices to strengthen first, especially when multiple sluices shared the same safety class.

3. Future research should explore fuzzy AHP to reduce subjectivity, conduct sensitivity analysis on the importance of index parameters, and extend the framework to include cost-benefit analysis. Validation in other river basins would also enhance generalizability.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares that they have no conflicts of interest.

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